STAGE 2 – PRACTICAL ENGINEERING CHALLENGE

TEAM ID:	TEAM NAME:
LRC-26-0570	ackerman.

Problem Statement: Industrial Manufacturing Process Automations

Industrial manufacturing depends on highly controlled liquid-handling systems to deliver process fluids reliably from a source or bulk supply to downstream process equipment. Depending on the application, these fluids may include **ultrapure fluids, process chemicals, solvents, developers, or CMP-related fluids**. Industry standards explicitly address liquid-chemical distribution, high-purity piping, bulk chemical distribution, system qualification, and UPW system operation and monitoring.

In such systems, **flow is a fundamental process variable**. An incorrect flow rate can indicate or cause a restriction, incorrect valve state, inadequate pumping performance, abnormal process demand, or another hydraulic deviation. The consequence may be unstable process conditions, incorrect chemical delivery, equipment malfunction, or loss of process performance.

This challenge therefore focuses on a fundamental industrial facilities and process-engineering task: **To establish a defined process-fluid flow, measure it reliably, detect a deviation, diagnose its cause, and restore the process to the required operating condition.**

Engineering Challenge

As you are part of a multidisciplinary engineering team responsible for designing, building and validating a small-scale automated liquid-flow system represented of industrial process handling system.

CASE 1 – CONTROL + MEASUREMNT
Problem: Develop a system that manages the changing level of fluid in the sump automatically (when there is a sudden external fluid charging to the sump) and while accurately measuring and totalizing the fluid discharged from the sump using the flow transmitter.

CASE 2 – FLUSHING +SHUTDOWN
Problem: Develop an alternative operating configuration that allows flush water to pass through the system while all three pumps operate simultaneously, and then safely transition the system to an OFF condition when the flush-water supply has been exhausted.

Given Process Parameters

Pipe diameter =	**8mm**
Fluid density (ρ) =	**1000kg/m^3**
Dynamic Viscosity (μ) =	**1x10^{-3}Pa.s**
Required Flow range =	**1-6 L/min**
Nominal target flow for the live challenge =	**3L/min**

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Available Hardware

Mechanical Hardware	Electronic Hardware
1. Water Tank – 3 Quantity 2. Pumps – 3 Quantity 3. FT-01 Flow Transmitter – 01 Quantity 4. Ball valve – 3 Quantity 5. ON/OFF/XV Valve – 1 Quantity 6. 10-mm OD tubing/blue – 9 meters/1 meter 7. Microcontroller/PCB/ data-acquisition software 8. Required electrical/ electronic accessories	1. Custom PCB 2. ESP32 NodeMCU module 3. Pump Driver-1 4. Pump Driver-2 5. Pump Driver-3 6. 12 V DC power input (**SMPS**) 7. 12 VDC Solenoid-valve interface 8. Flow-sensor interface 9. Float-switch interface 10. ON/OFF-switch interface 11. Pump-driver PWM/enable interfaces 12. On/Off Switch

The team must determine the appropriate arrangement, connections and measurement method.

- Assemble and commission the liquid-flow system using the components provided.
- Measure and characterize the flow using FT-01
- Interface FT-01 with the provided **data-acquisition system**
- Experimentally determine the relationship between sensor output and actual flow.
- Characterize the effect of valve position on flow rate. Record and analyse flow vs time data
- Maintain target flow of 3L/min
- Use graphs and measured data to justify engineering decisions
- Design an appropriate hydraulic arrangement and P&ID

TEAM INSTRUCTIONS

- As a team go through the list provided to read about each component that has been Listed.
- The PCB is a 12 V DC and the SMPS connections should be made carefully.

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TASK 1: HYDRAULICS SYSTEM DESIGN

Q1.1 Design and build the hydraulic path with the flow transmitter on the recirculation line. Submit/sketch a labelled P&ID below by showing the pumps, Needle valves, flow transmitter, flow direction, Tank, sump, tube diameter and measurement points.

CASE 1 – Process Fluid Transfer & Automatic Level Protection

Objective: Set-up and operate a closed-loop process- fluid handling system using the available tank, pump, valves flow transmitter, sump, float sensor and control PCB. The System must continuously transfer process fluid recirculation system by connecting the 10L fluid tank with a On/Off valve to a pump-01 and the pump-01 goes to the inlet of sump 4.2L tank. The Sump 4.2L outlet line is connected to another pump-02 followed by a flow transmitter that measures the total flow back to the 10L fluid tank. The Pump-01 runs on 50% duty cycle and Pump-02 runs on 100% duty cycle. **Calibrate the flow transmitter** and demonstrate correct fluid circulation, if there is sudden introduction of external fluid into the Sump tank then Pump-01 should stop while pump – 02 should still runs till fluid level reduces in the sump. After the fluid reduction in the SUMP 4.2L tank the Pump-01 should turn on and the process must continue. **Also, demonstrate Flow vs Time trends in mL/s, total transferred Liters back to the 10L tank** (share Screenshots flow in L/min). **Share the code.** The case-1 can be turned on using the available on/off Switch, to start the system.

CASE 2 – Flush/Line Conditioning Sequence

Objective: After successfully completing case 1, initiate the flush water in the sequence using the **equal Tee.** The On/Off valve must be switched off in order to isolate the 10L process fluid tank. The flush Fluid tank (4.2L) is used as a temporary fluid source to flush out the process fluid or any debris from the system. Before the start of this process make sure you open the manual ball valve on the fluid flush line before Pump-03 starts. The entire process of case -1 and case - 2 should run in a single go without manual intervention except opening of ball valve before the Pump-03 starts. **The System must stop as soon as the tank flush fluid 4.2L tank finishes.** The system must be automated without human interventions (only ball valve can be opened manually before the start of flush line sequence)

SUBMIT:

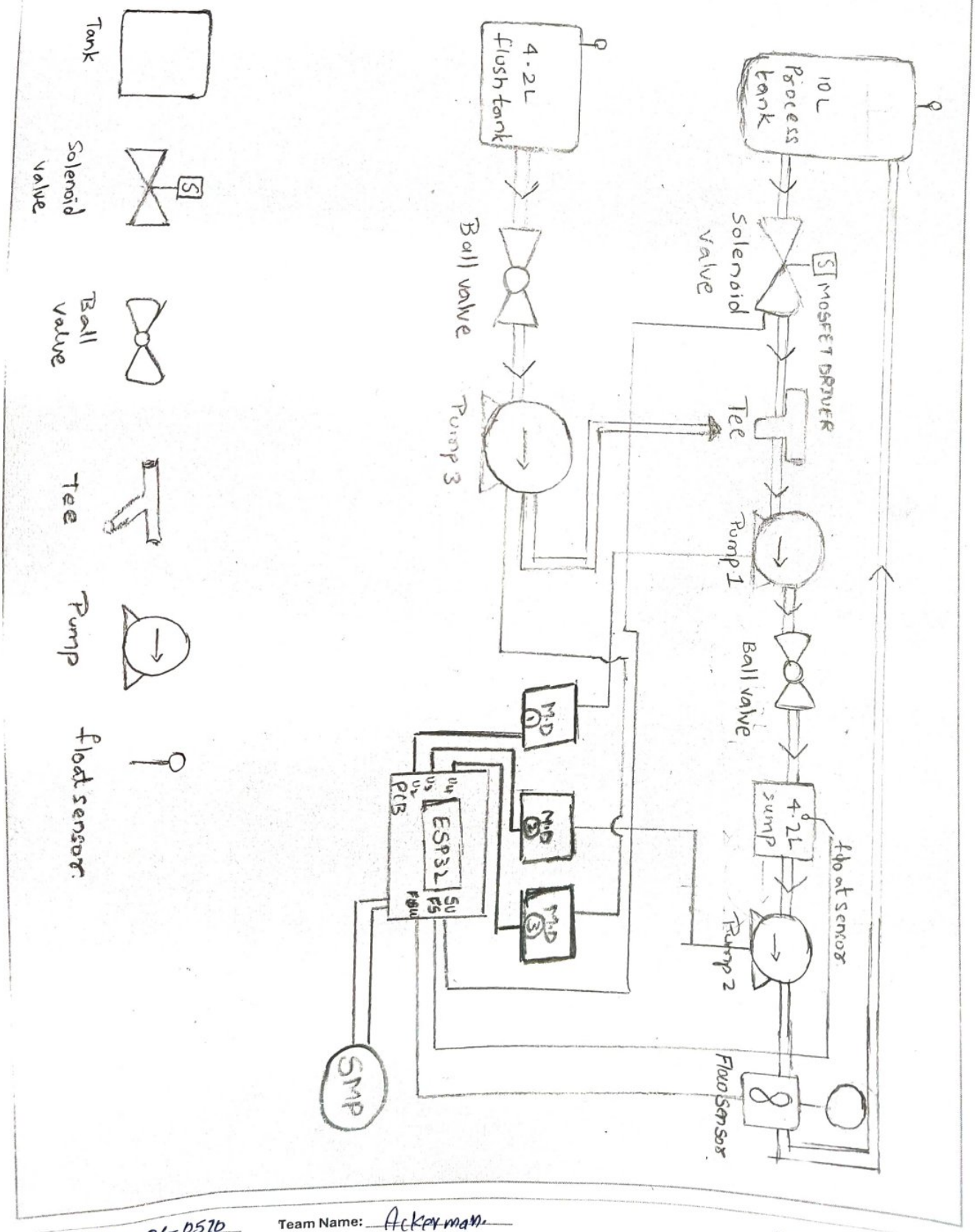
A. Submit: 10-15 min Video and Photograph of the completed setup, with identification of each component and explain the direction of flow in the video verbally. Demonstrate both Case - 1 and Case – 2.

B. Final P&ID Block diagram

P.T.O →

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P&ID SKETCH / WORKING SPACE



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Q1.2 After setting-up the system, now operate it at approximately at 1L/min, 3L/min & 6L/min. Compare the theoretical velocity with the actual operating behaviour.

Theoretical calculation:

$Q = Av$ | v = fluid velocity (m/s)
$v = \frac{Q}{A}$ | Q = volumetric flow (m^3/s)
A = Area (m^2)
D = diameter = 0.008m = 8mm

Step 1:

$$A = \frac{\pi D^2}{4}$$

$$A = \frac{\pi (0.008)^2}{4} = 5.0265 \times 10^{-5}\ m^2$$

Step 2:

$1L = 10^{-3}\ m^3$

$$Q(m^3/s) = \frac{Q(L/min) \times 10^{-3}}{60}$$

Step 3:

For 1L/min:

$$v = \frac{(1 \times 10^{-3})/60}{5.0265 \times 10^{-5}} = 0.332\ m/s$$

For 3L/min:

$$v = \frac{(3 \times 10^{-3})/60}{5.0265 \times 10^{-5}} = 0.995\ m/s$$

For 6L/min:

$$v = \frac{(6 \times 10^{-3})/60}{5.0265 \times 10^{-5}} = 1.989\ m/s.$$

Practical values:

At 1.017 L/min: $v = 0.338$ m/s

At 2.727 L/min: $v = 0.905$ m/s.

★ The 6L/min velocity cannot be calculated because that operating flow rate was not achieved.

- Theoritically it touched 1.989m/s with 8mm tube but not in practical.

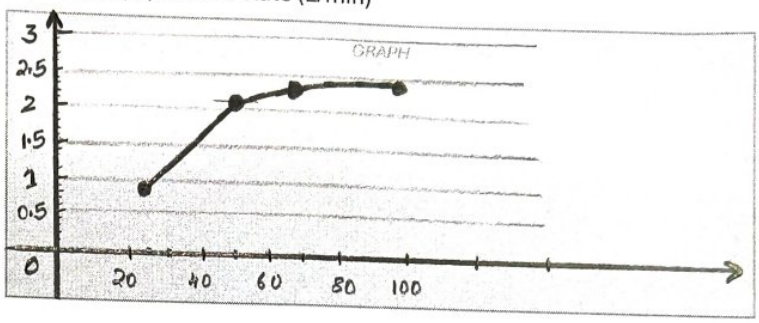
Theoretical flow	T. velocity	Actual flow	A. velocity.
1L	0.332	1.017	0.338
3L	0.995	2.727	0.907
6L	1.989	Not achieved	Not determined

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TASK 2: FLOW RESPONSE TO VALVE POSITION

Q 2.1 Set the ball valve to approximately 25%,50%,75% & 100%. At each position, allow the flow to stabilize and record the flow rate. Create the following graph of valve position (%) vs Flow Rate (L/min)

VALVE POSITION	FLOW RATE
25%	0.849 L/min
50%	2.066 L/min
70%	2.296 L/min
100%	2.388 L/min



Q2.2 Determine the experimental calibration equation: $Q = af + b$, where Q = flow rate in L/min, f = pulse frequency in Hz, a = experimental calibration coefficient and b = offset.

Report: $Q = af + b = 0.010089f - 0.012031$

R^2 =	0.99934
a =	0.010089
b =	- 0.012031
Maximum calibration error =	0.0333 L/m / 1.47 %
Average calibration error =	0.01632 l/min / 0.90 %

Q2.3 FT-01 produces pluses corresponding to water flow. Run the system at least for 5 different flow conditions within the available operating range (Hint: use formulas f=N/t and Q=V/t). For each condition, record the following:

Trial	Pulses	Time(s)	Frequency (Hz)	Collected Volume (L)	Actual Flow (L/min)
1	5,979	117	51.10	1.060	0.513
2	6,045	59	102.46	1.028~	1.017
3	6,007	33	182.03	1.0	1.818
4	6,055	26	232.88	1.0	2.308
5	5,867	22	266.68	1.0	2.727

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CALCULATIONS / OBSERVATIONS

$Q_{actual} = \frac{V \times 60}{t}$

- Trail (1) – $f = \frac{N}{t} = \frac{5,979}{117} = 51.10256$
- Trail (2) – $f = \frac{N}{t} = \frac{6,045}{59} = 102.457$
- Trail (3) – $f = \frac{N}{t} = \frac{6007}{33} = 182.03030$
- Trail (4) – $f = \frac{N}{t} = \frac{6055}{26} = 232.8846$
- Trail (5) – $f = \frac{N}{t} = \frac{5,867}{22} = 266.68181$

→ The volume in the 2.3 question is collected and measured carefully to get precise calculation., we measured all the readings using a 1L Beaker along with a 100ml measuring jal.

→ No Readings were Synthetic, all values are taken manually in each cycle

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TASK 3: HYDRAULIC SYSTEM THEORY AND FAULT ANALYSIS

Q3.1 The effective liquid volume inside a section of tubing is 0.5L, while the operating flow rate is 2.5L/min. The hydraulic residence time is given by $T_r = Q/V$. Calculate the residence time in seconds, also answer the following questions:

1. What does residence time physically represent in this system?
2. What happens to residence time if the tubing volume is doubled while the flow rate remains constant?
3. What happens to residence time if the flow rate is doubled while the tubing volume remains constant?
4. Why is residence time important when interpreting a sudden change in measured flow?

Given $V=0.5L$ Flowrate $Q=2.5$ L/min $T_r=Q/V$

$T_r = \frac{0.5}{2.5} = 0.2$ minutes $\times 60 = 12$ seconds

1. Residence time represents the avg time taken by the liquid to pass through the tubing section containing 0.5L of liquid at flowrate of 2.5 L/min.

2. Residence time is directly proportional to the tubing volume is doubles while the flow rate remains const. If tubing volume is doubled from 0.5L to 1.0L, while flow rate remains 2.5 L/min, the residence time also doubles.
$T_r = 1.0/2.5 * 60 = 24$ seconds

3. Residence time is inversely proportional to the flow rate. If flowrate increases from 2.5L/min to 5L/min while tubing volume remains at 0.5L, residence time decreases by half.
$T_r = 0.5/5 \times 60 = 6$ seconds

4. Residence time helps us understand the delay between a change in the flowrate and its effect at a downstream measurement point, When the pump speed or valve position changes, the liquid already present inside the tubing continues to move before the new operating condition is fully reflected at the measurement point. Understanding this delay helps distinguish a genuine flow change from a temporary response caused by liquid ~~involvement~~ movement through the tubing. It is therefore useful for monitoring hydralic performance and identifying abnormal operating conditions.

Conclusion!
The calculated residence time is 12 seconds. It increases when the tubing volume increases and decreases when the flow rate increases.

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Q3.2 The following observation are obtained from the same system:

CONDITION	PUMP	VALVE	FT-01 PULSE SIGNAL	MEASURED FLOW
A	ON	Open	Normal	Normal
B	ON	Partially closed	Reduced	Reduced
C	ON	Open	Zero	Zero
D	OFF	Open	Zero	Zero
E	ON	Open	Intermittent	Fluctuating

For each of the condition, identify the most likely physical or instrumentation condition. Then answer: Which condition is most difficult to distinguish using FT-01 alone, and what additional information would you require to distinguish the fault? (Hint: Teams must consider possibilities such as: Valve restriction, pump failure, hydraulic blockage, sensor/signal failure, unstable flow etc)

condition	observation	Likely explanation
A.	Pump on, valve OPEN, normal pulses & flow.	Normal operation.
B.	Pump ON, valve partly closed, reduced pulses & flow	Valve restriction
C.	Pump ON, valve open, zero pulses & zero flow	Blockage or Pump failure or sensor failure.
D.	Pump off, valve open, zero pulses & flow	Normal stopped condition.
E.	Pump on, valve open, intermitten pulses & fluctuating flow	Air in line, irregular pump, loose wiring or faulty sensor

The condition which is most difficult to distinguish using FT-01 alone is condition C, because zero pulses & zero flow look identical whether the cause is a hydraulic blockage, pump failure or transmitter failure.

Additional information required:
1. Pump status / current: It confirms the pump is actually running.
2. Pressure measurement: It detects a blockage or restriction.
3. Independent flow measurement: Shows if liquid flows when FT-01 reads zero.
4. Wiring / signal inspection: finds loose connections or transmitter failure.

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HARDWARE LIST

Mechanical Hardware

Component	Description	Main Use
Ball Valve (BV-01)	A manually operated valve with a spherical shut-off mechanism.	Used to **manually isolate or permit fluid flow** in the pipeline. It is useful for maintenance, testing, and flow isolation.
10-Litre Fluid Tank (Fluid Tank-01)	Transparent storage tank with approximately 10 L capacity.	Used as the **main fluid reservoir** for storing process fluid. The transparent body also allows the fluid level and condition to be visually monitored.
5-Litre Sump	Small transparent reservoir labelled **SUMP**, approximately 4.5 L.	Used as a **collection/storage vessel** for fluid returning to or entering the sump section of the system.
4.2-Litre Flush Fluid Tank / Flush Line	The transparent vessel labelled **FLUSH FLUID** is approximately 4.5 L.	Provides **flush fluid** for cleaning or flushing the main pipe circuit when required.
ON/OFF Valve (OV-01)	Electrically operated valve/solenoid-type valve shown in the setup.	Used for **automatic control of fluid flow**, allowing the controller to turn a particular flow path on or off.
Pumps -Pump-1, Pump-2, Pump-3	Motor-driven fluid pumps. Three pumps are shown in the setup.	Used to **circulate and transfer fluid** between different sections of the system. Each pump can be independently controlled.
8 mm Process Pipe	Flexible blue tubing with approximately 8 mm process-line size.	Used as the **main fluid-flow path** connecting tanks, valves, pumps, tees, and other components.
10 mm × 8 mm Reducers	Reducer fittings that transition between a 10 mm connection and an 8 mm pipe.	Used to **adapt the pump connection size to the 8 mm process pipe. Mainly used for Pumps**
8 mm Connectors	Push-fit/barbed-type fittings compatible with the 8 mm tubing.	Used to **join pipes to components** and to make detachable connections between sections of tubing.
8 mm Equal Tee	T-shaped connector having three equal 8 mm ports.	Used to **split one 8 mm fluid line into two branches or combine two branches into one line.**

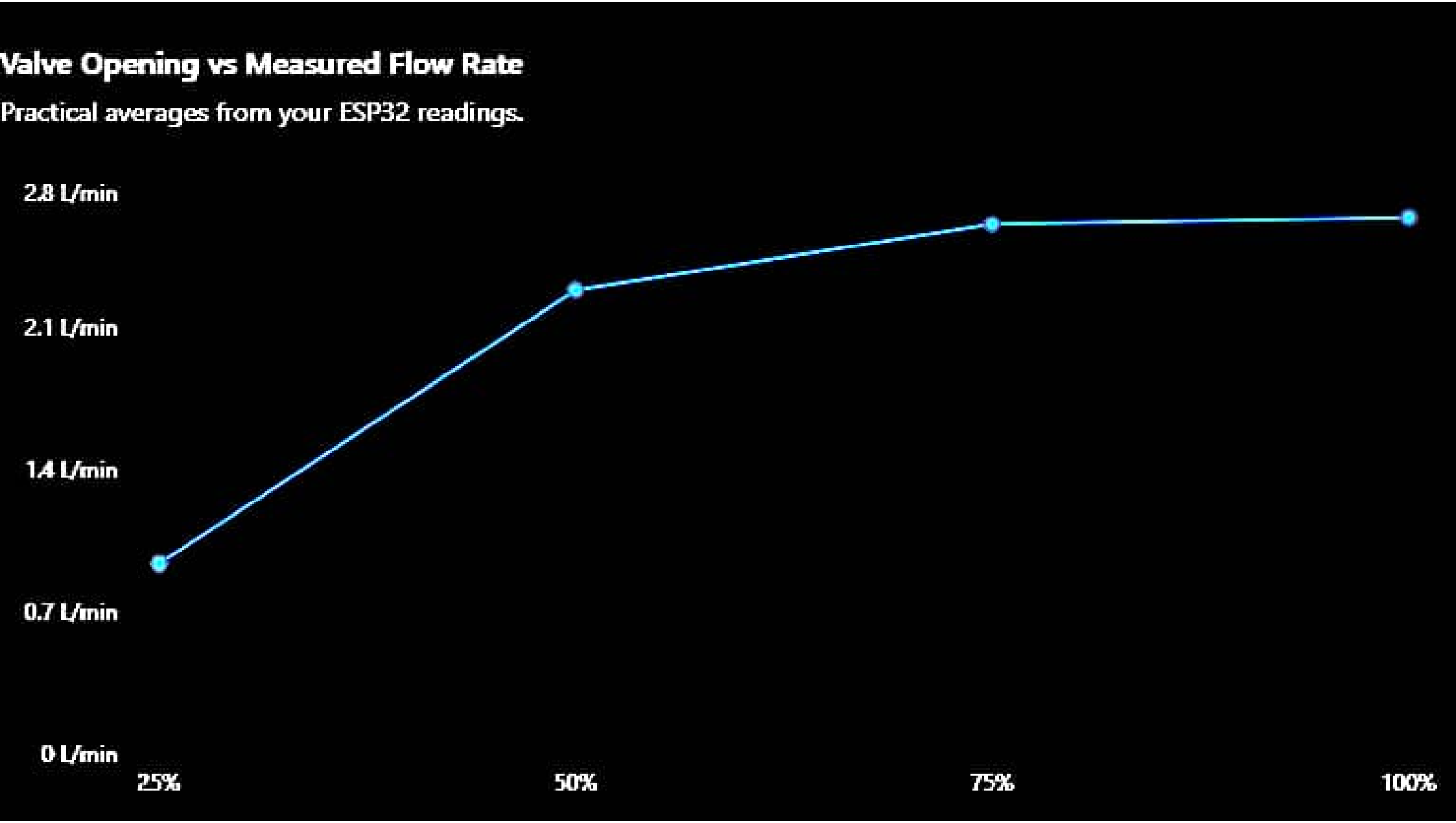
Electronic Hardware

Electronic Component	Identification / Function	Use
PCB -Printed Circuit Board	Custom control board	Provides the electrical interconnection between the **12 V power supply, ESP32 controller, sensors, valves, indicators, and pump-driver interfaces.**
ESP32 / NodeMCU ESP32S module (U7)	Main microcontroller	Acts as the **central controller.** It provides GPIO/PWM signals for controlling outputs and receiving signals from sensors/switches.

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Electronic Component	Identification / Function	Use
12 V DC Input Connector	Marked **12V INPUT**	Connection point for the external 12 V DC power supply.
12 V Output Connectors	P1, P2, P3 and additional 12 V output terminals	Provide 12 V power connections for external loads such as the pump-driver/motor circuits.
Motor Driver Interfaces -U2, U3, U4	Three multi-pin driver connectors shown on the PCB	Provide dedicated interfaces for **Pump-1, Pump-2 and Pump-3 driver/control connections.**
Solenoid Valve Connector	Labelled **SOLENOID_VALVE -S/V/G**	Interface for controlling/connecting the solenoid or ON/OFF valve.
Flow Sensor Connector	Labelled **FLOW_SENSOR - S/V/G**	Interface for the water-flow sensor; supplies power and receives the sensor signal. **THE FLOW SENSOR SHOULD BE CALIBRATED**
Float Switch Connector	Labelled **FLOAT_SWITCH -FSW/GND**	Interface for a float/level switch used to detect a water-level condition.
ON/OFF Switch Connector	Labelled **ON/OFF_SWITCH -SW/GND**	Interface for an external manual ON/OFF switch.
12V DC SMPS	**12V DC SMPS**	For Power Supply to the board

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Code

```
#include <Arduino.h>

// ---------- FLOW SENSOR ----------

#define FLOW_SENSOR_PIN 14

const float PULSES_PER_LITRE = 6207.0f;

const float FLUSH_TARGET_L = 3.2f;

// ---------- INPUTS ----------

#define START_SWITCH_PIN 34

#define FLOAT_SENSOR_PIN 35

// ---------- SOLENOID ----------

#define SOLENOID_PIN 27

// Assumption: HIGH = OPEN, LOW = CLOSED

// ---------- PUMP 1: DRIVER U2 ----------

#define P1_RPWM 33

#define P1_LPWM 32

#define P1_REN  26

#define P1_LEN  13

// ---------- PUMP 2: DRIVER U3 ----------

#define P2_RPWM 18

#define P2_LPWM 5

#define P2_REN  17

#define P2_LEN  16

// ---------- PUMP 3: DRIVER U4 ----------

#define P3_RPWM 25

#define P3_LPWM 23

#define P3_REN  22

#define P3_LEN  21
```

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```
const int PWM_FREQ = 20000;
const int PWM_RESOLUTION = 8;
const int PUMP1_DUTY = 128; // approximately 50%
const int PUMP2_DUTY = 255; // 100%
const int PUMP3_DUTY = 255; // 100%
// GPIO35 LOW = sump HIGH; HIGH = sump LOW
const int SUMP_HIGH_LEVEL = LOW;
const int SUMP_LOW_LEVEL = HIGH;
enum SystemState {
  WAITING_FOR_START,
  CASE1_RUNNING,
  CASE2_FLUSHING,
  SYSTEM_COMPLETE,
  SYSTEM_ERROR};
SystemState state = WAITING_FOR_START;
volatile uint32_t totalPulses = 0;
volatile uint32_t intervalPulses = 0;
portMUX_TYPE flowMux = portMUX_INITIALIZER_UNLOCKED;
bool sumpHighSeen = false;
bool case2Started = false;
bool startArmed = false;
uint32_t lastStartChange = 0;
uint32_t lastTelemetry = 0;
uint32_t lastFloatChange = 0;
int lastStartRaw = HIGH;
int stableStart = HIGH;
int lastFloatRaw = HIGH;
int stableFloat = HIGH;
```

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```
uint32_t flushStartPulses = 0;
// ---------- FLOW SENSOR ISR ----------
void IRAM_ATTR flowPulseISR() {
  portENTER_CRITICAL_ISR(&flowMux);
  totalPulses++;
  intervalPulses++;
  portEXIT_CRITICAL_ISR(&flowMux);}
// ---------- PUMP CONTROL ----------
bool attachPWM(int pin) {
  return ledcAttach(pin, PWM_FREQ, PWM_RESOLUTION);}
void setPump(int rpwm, int lpwm, int ren, int len, int duty) {
  if (duty <= 0) {
    ledcWrite(rpwm, 0);
    ledcWrite(lpwm, 0);
    digitalWrite(ren, LOW);
    digitalWrite(len, LOW);
    return;}
  digitalWrite(ren, HIGH);
  digitalWrite(len, HIGH);
  ledcWrite(lpwm, 0);
  ledcWrite(rpwm, duty);}
void pump1On() {
  setPump(P1_RPWM, P1_LPWM, P1_REN, P1_LEN, PUMP1_DUTY);}
void pump1Off() {
  setPump(P1_RPWM, P1_LPWM, P1_REN, P1_LEN, 0);}
void pump2On() {
  setPump(P2_RPWM, P2_LPWM, P2_REN, P2_LEN, PUMP2_DUTY);}
void pump2Off() {
```

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```
  setPump(P2_RPWM, P2_LPWM, P2_REN, P2_LEN, 0);}
void pump3On() {
  setPump(P3_RPWM, P3_LPWM, P3_REN, P3_LEN, PUMP3_DUTY);}
void pump3Off() {
  setPump(P3_RPWM, P3_LPWM, P3_REN, P3_LEN, 0);}
void stopAllPumps() {
  pump1Off();
  pump2Off();
  pump3Off();}
uint32_t getTotalPulses() {
  portENTER_CRITICAL(&flowMux);
  uint32_t value = totalPulses;
  portEXIT_CRITICAL(&flowMux);
  return value;}
uint32_t takeIntervalPulses() {
  portENTER_CRITICAL(&flowMux);
  uint32_t value = intervalPulses;
  intervalPulses = 0;
  portEXIT_CRITICAL(&flowMux);
  return value;}
void resetFlowCounters() {
  portENTER_CRITICAL(&flowMux);
  totalPulses = 0;
  intervalPulses = 0;
  portEXIT_CRITICAL(&flowMux);}
void startCase2() {
  if (case2Started) return;
  case2Started = true;
```

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```
  state = CASE2_FLUSHING;
  // Close solenoid and start Pump 3 immediately.
  digitalWrite(SOLENOID_PIN, LOW);
  flushStartPulses = getTotalPulses();
  pump3On();
  Serial.println("\nCASE 2 STARTED");
  Serial.println("Solenoid: CLOSED");
  Serial.println("Pump 3: ON at 100%");
  Serial.println("Pump 1 and Pump 2 continue Case 1 control.");}
void finishSystem() {
  stopAllPumps();
  digitalWrite(SOLENOID_PIN, LOW);
  state = SYSTEM_COMPLETE;
  Serial.println("\nTARGET REACHED: 3.2 L");
  Serial.println("All pumps OFF.");
  Serial.println("Solenoid CLOSED.");
  Serial.println("SYSTEM COMPLETE.");}
void setup() {
  Serial.begin(115200);
  pinMode(START_SWITCH_PIN, INPUT); // External pull-up required
  pinMode(FLOAT_SENSOR_PIN, INPUT); // External bias required
  pinMode(FLOW_SENSOR_PIN, INPUT_PULLUP);
  pinMode(SOLENOID_PIN, OUTPUT);
  pinMode(P1_REN, OUTPUT);
  pinMode(P1_LEN, OUTPUT);
  pinMode(P2_REN, OUTPUT);
  pinMode(P2_LEN, OUTPUT);
  pinMode(P3_REN, OUTPUT);
```

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```
  pinMode(P3_LEN, OUTPUT);
  digitalWrite(SOLENOID_PIN, LOW);
  stopAllPumps();
  bool pwmOK =
    attachPWM(P1_RPWM) && attachPWM(P1_LPWM) &&
    attachPWM(P2_RPWM) && attachPWM(P2_LPWM) &&
    attachPWM(P3_RPWM) && attachPWM(P3_LPWM);
  if (!pwmOK) {
    stopAllPumps();
    state = SYSTEM_ERROR;
    Serial.println("ERROR: PWM setup failed. Check pins and ESP32 core.");
    return;}
  attachInterrupt(
    digitalPinToInterrupt(FLOW_SENSOR_PIN),
    flowPulseISR,
    RISING);
  // Require the start switch to be released before arming.
  lastStartRaw = digitalRead(START_SWITCH_PIN);
  stableStart = lastStartRaw;
  startArmed = (stableStart == HIGH);
  lastFloatRaw = digitalRead(FLOAT_SENSOR_PIN);
  stableFloat = lastFloatRaw;
  Serial.println("\nSYSTEM READY");
  Serial.println("Release the start switch, then press it to begin.");}
void loop() {
  if (state == SYSTEM_ERROR || state == SYSTEM_COMPLETE) {
  return;}
  uint32_t now = millis();
```

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```
// ---------- START SWITCH DEBOUNCE ----------
int startRaw = digitalRead(START_SWITCH_PIN);
if (startRaw != lastStartRaw) {
  lastStartRaw = startRaw;
  lastStartChange = now;}
if (now - lastStartChange >= 40) {
  if (stableStart != startRaw) {
    stableStart = startRaw;
    if (stableStart == HIGH) {
      startArmed = true;
    } else if (startArmed && state == WAITING_FOR_START) {
      startArmed = false;
      resetFlowCounters();
      sumpHighSeen = false;
      case2Started = false;
      state = CASE1_RUNNING;
      digitalWrite(SOLENOID_PIN, HIGH); // Open valve
      pump1On();
      pump2On();
      pump3Off();
      Serial.println("\nCASE 1 STARTED");
      Serial.println("Solenoid OPEN, Pump 1 at 50%, Pump 2 at 100%.");}}}
if (state != CASE1_RUNNING && state != CASE2_FLUSHING) {
  return;}
// ---------- FLOAT SENSOR DEBOUNCE ----------
int floatRaw = digitalRead(FLOAT_SENSOR_PIN);
if (floatRaw != lastFloatRaw) {
  lastFloatRaw = floatRaw;
```

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```
  lastFloatChange = now;}
if (now - lastFloatChange >= 50) {
  if (stableFloat != floatRaw) {
    stableFloat = floatRaw;
  }}
// ---------- CASE 1 SUMP-LEVEL CONTROL ----------
if (stableFloat == SUMP_HIGH_LEVEL) {
  sumpHighSeen = true;
  pump1Off();  // Stop filling the sump
  pump2On();   // Continue draining the sump
} else if (stableFloat == SUMP_LOW_LEVEL) {
  pump1On();   // Resume Pump 1 immediately
  pump2On();
  // Start Case 2 on the first return to LOW after HIGH.
  if (sumpHighSeen && !case2Started) {
    startCase2();}}
// ---------- CASE 2: MEASURE FLUSH VOLUME ----------
if (state == CASE2_FLUSHING) {
  uint32_t currentPulses = getTotalPulses();
  uint32_t flushPulses = currentPulses - flushStartPulses;
  float flushLitres =
    (float)flushPulses / PULSES_PER_LITRE;
  if (flushLitres >= FLUSH_TARGET_L) {
    finishSystem();
    return;
  }
}
// ---------- TELEMETRY ----------
```

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```
  if (now - lastTelemetry >= 1000) {
    float elapsedSeconds = (now - lastTelemetry) / 1000.0f;
    lastTelemetry = now;
    uint32_t pulsesThisInterval = takeIntervalPulses();
    float flowHz = pulsesThisInterval / elapsedSeconds;
    float flowLMin = flowHz * 60.0f / PULSES_PER_LITRE;
    float flowMlSec = flowHz * 1000.0f / PULSES_PER_LITRE;
    float totalLitres =
      (float)getTotalPulses() / PULSES_PER_LITRE;
    float flushLitres = 0.0f;
    if (case2Started) {
      flushLitres =
        (float)(getTotalPulses() - flushStartPulses)
        / PULSES_PER_LITRE;
    }
    Serial.printf(
      "Stage:%s | SumpHigh:%s | Flow:%.3f L/min | "
      "Total:%.3f L | Flush:%.3f / %.1f L\n",
      state == CASE2_FLUSHING ? "CASE 2" : "CASE 1",
      stableFloat == SUMP_HIGH_LEVEL ? "YES" : "NO",
      flowLMin,
      totalLitres,
      flushLitres,
      FLUSH_TARGET_L);
  }}
```

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25% Percent

```
04:46:00.571 -> System: STOPPED | Float: HIGH LEVEL | Pump-01: OFF
04:46:46.651 -> HIGH LEVEL: Pump-02 ON; Pump-01 OFF.
04:46:47.619 -> Flow: 0.816 L/min | 13.60 mL/s | Total: 0.014 L | Pump-01: OFF
04:46:48.652 -> Flow: 0.965 L/min | 16.08 mL/s | Total: 0.030 L | Pump-01: OFF
04:46:49.617 -> Flow: 0.940 L/min | 15.66 mL/s | Total: 0.045 L | Pump-01: OFF
04:46:50.635 -> Flow: 0.940 L/min | 15.66 mL/s | Total: 0.061 L | Pump-01: OFF
04:46:51.635 -> Flow: 0.940 L/min | 15.66 mL/s | Total: 0.077 L | Pump-01: OFF
04:46:52.634 -> Flow: 0.965 L/min | 16.08 mL/s | Total: 0.093 L | Pump-01: OFF
04:46:53.617 -> Flow: 0.940 L/min | 15.66 mL/s | Total: 0.108 L | Pump-01: OFF
04:46:54.618 -> Flow: 0.940 L/min | 15.66 mL/s | Total: 0.124 L | Pump-01: OFF
04:46:55.618 -> Flow: 0.940 L/min | 15.66 mL/s | Total: 0.140 L | Pump-01: OFF
04:46:56.634 -> Flow: 0.940 L/min | 15.66 mL/s | Total: 0.155 L | Pump-01: OFF
04:46:57.618 -> Flow: 0.965 L/min | 16.08 mL/s | Total: 0.171 L | Pump-01: OFF
04:46:58.617 -> Flow: 0.940 L/min | 15.66 mL/s | Total: 0.187 L | Pump-01: OFF
04:46:59.634 -> Flow: 0.940 L/min | 15.66 mL/s | Total: 0.203 L | Pump-01: OFF
04:47:00.635 -> Flow: 0.940 L/min | 15.66 mL/s | Total: 0.218 L | Pump-01: OFF
04:47:01.618 -> Flow: 0.940 L/min | 15.66 mL/s | Total: 0.234 L | Pump-01: OFF
04:47:02.634 -> Flow: 0.915 L/min | 15.25 mL/s | Total: 0.249 L | Pump-01: OFF
04:47:03.635 -> Flow: 0.940 L/min | 15.66 mL/s | Total: 0.265 L | Pump-01: OFF
04:47:04.634 -> Flow: 0.940 L/min | 15.66 mL/s | Total: 0.281 L | Pump-01: OFF
04:47:05.634 -> Flow: 0.940 L/min | 15.66 mL/s | Total: 0.296 L | Pump-01: OFF
04:47:06.634 -> Flow: 0.940 L/min | 15.66 mL/s | Total: 0.312 L | Pump-01: OFF
04:47:07.617 -> Flow: 0.940 L/min | 15.66 mL/s | Total: 0.328 L | Pump-01: OFF
04:47:08.633 -> Flow: 0.940 L/min | 15.66 mL/s | Total: 0.343 L | Pump-01: OFF
04:47:09.617 -> Flow: 0.940 L/min | 15.66 mL/s | Total: 0.359 L | Pump-01: OFF
04:47:10.618 -> Flow: 0.940 L/min | 15.66 mL/s | Total: 0.375 L | Pump-01: OFF
04:47:11.031 -> SYSTEM STOPPED
```

50% PERCENT

```
04:48:25.496 -> System: STOPPED | Float: HIGH LEVEL | Pump-01: OFF
04:48:29.646 -> HIGH LEVEL: Pump-02 ON; Pump-01 OFF.
04:48:30.611 -> Flow: 2.152 L/min | 35.86 mL/s | Total: 0.036 L | Pump-01: OFF
04:48:31.612 -> Flow: 2.325 L/min | 38.75 mL/s | Total: 0.075 L | Pump-01: OFF
04:48:32.612 -> Flow: 2.300 L/min | 38.34 mL/s | Total: 0.113 L | Pump-01: OFF
04:48:33.611 -> Flow: 2.300 L/min | 38.34 mL/s | Total: 0.151 L | Pump-01: OFF
04:48:34.629 -> Flow: 2.325 L/min | 38.75 mL/s | Total: 0.190 L | Pump-01: OFF
04:48:35.612 -> Flow: 2.300 L/min | 38.34 mL/s | Total: 0.228 L | Pump-01: OFF
04:48:36.611 -> Flow: 2.300 L/min | 38.34 mL/s | Total: 0.267 L | Pump-01: OFF
04:48:37.611 -> Flow: 2.325 L/min | 38.75 mL/s | Total: 0.305 L | Pump-01: OFF
04:48:38.643 -> LEVEL NORMAL: Pump-01 restarted; Pump-02 ON.
04:48:38.643 -> Flow: 2.292 L/min | 38.21 mL/s | Total: 0.345 L | Pump-01: ON
04:48:39.644 -> Flow: 2.300 L/min | 38.34 mL/s | Total: 0.383 L | Pump-01: ON
04:48:40.644 -> Flow: 2.300 L/min | 38.34 mL/s | Total: 0.421 L | Pump-01: ON
04:48:41.644 -> Flow: 2.276 L/min | 37.93 mL/s | Total: 0.459 L | Pump-01: ON
04:48:42.662 -> Flow: 2.300 L/min | 38.34 mL/s | Total: 0.498 L | Pump-01: ON
04:48:43.644 -> Flow: 2.276 L/min | 37.93 mL/s | Total: 0.535 L | Pump-01: ON
04:48:44.644 -> Flow: 2.300 L/min | 38.34 mL/s | Total: 0.574 L | Pump-01: ON
04:48:45.644 -> Flow: 2.300 L/min | 38.34 mL/s | Total: 0.612 L | Pump-01: ON
04:48:46.643 -> Flow: 2.276 L/min | 37.93 mL/s | Total: 0.650 L | Pump-01: ON
04:48:47.660 -> Flow: 2.300 L/min | 38.34 mL/s | Total: 0.688 L | Pump-01: ON
04:48:48.644 -> Flow: 2.276 L/min | 37.93 mL/s | Total: 0.726 L | Pump-01: ON
04:48:49.643 -> Flow: 2.300 L/min | 38.34 mL/s | Total: 0.765 L | Pump-01: ON
04:48:50.660 -> Flow: 2.276 L/min | 37.93 mL/s | Total: 0.803 L | Pump-01: ON
04:48:51.642 -> Flow: 2.300 L/min | 38.34 mL/s | Total: 0.841 L | Pump-01: ON
04:48:52.626 -> Flow: 2.276 L/min | 37.93 mL/s | Total: 0.879 L | Pump-01: ON
04:48:53.643 -> Flow: 2.300 L/min | 38.34 mL/s | Total: 0.917 L | Pump-01: ON
04:48:54.644 -> Flow: 2.276 L/min | 37.93 mL/s | Total: 0.955 L | Pump-01: ON
04:48:55.660 -> Flow: 2.276 L/min | 37.93 mL/s | Total: 0.993 L | Pump-01: ON
04:48:56.660 -> Flow: 2.276 L/min | 37.93 mL/s | Total: 1.031 L | Pump-01: ON
04:48:57.642 -> Flow: 2.276 L/min | 37.93 mL/s | Total: 1.069 L | Pump-01: ON
```

75% PERCENT

```
04:52:43.710 -> HIGH LEVEL: Pump-02 ON; Pump-01 OFF.
04:52:44.694 -> Flow: 2.473 L/min | 41.22 mL/s | Total: 0.041 L | Pump-01: OFF
04:52:45.695 -> Flow: 2.622 L/min | 43.70 mL/s | Total: 0.085 L | Pump-01: OFF
04:52:46.710 -> Flow: 2.622 L/min | 43.70 mL/s | Total: 0.129 L | Pump-01: OFF
04:52:47.693 -> Flow: 2.622 L/min | 43.70 mL/s | Total: 0.172 L | Pump-01: OFF
04:52:48.693 -> Flow: 2.622 L/min | 43.70 mL/s | Total: 0.216 L | Pump-01: OFF
04:52:49.710 -> Flow: 2.622 L/min | 43.70 mL/s | Total: 0.260 L | Pump-01: OFF
04:52:50.694 -> Flow: 2.622 L/min | 43.70 mL/s | Total: 0.303 L | Pump-01: OFF
04:52:51.709 -> Flow: 2.622 L/min | 43.70 mL/s | Total: 0.347 L | Pump-01: OFF
04:52:52.027 -> LEVEL NORMAL: Pump-01 restarted; Pump-02 ON.
04:52:52.710 -> Flow: 2.597 L/min | 43.28 mL/s | Total: 0.390 L | Pump-01: ON
04:52:53.709 -> Flow: 2.622 L/min | 43.70 mL/s | Total: 0.434 L | Pump-01: ON
04:52:54.710 -> Flow: 2.597 L/min | 43.28 mL/s | Total: 0.477 L | Pump-01: ON
04:52:55.710 -> Flow: 2.622 L/min | 43.70 mL/s | Total: 0.521 L | Pump-01: ON
04:52:56.693 -> Flow: 2.597 L/min | 43.28 mL/s | Total: 0.564 L | Pump-01: ON
04:52:57.711 -> Flow: 2.622 L/min | 43.70 mL/s | Total: 0.608 L | Pump-01: ON
04:52:58.693 -> Flow: 2.622 L/min | 43.70 mL/s | Total: 0.652 L | Pump-01: ON
04:52:59.696 -> Flow: 2.597 L/min | 43.28 mL/s | Total: 0.695 L | Pump-01: ON
04:53:00.710 -> Flow: 2.622 L/min | 43.70 mL/s | Total: 0.739 L | Pump-01: ON
04:53:01.693 -> Flow: 2.622 L/min | 43.70 mL/s | Total: 0.782 L | Pump-01: ON
04:53:02.709 -> Flow: 2.597 L/min | 43.28 mL/s | Total: 0.826 L | Pump-01: ON
04:53:03.711 -> Flow: 2.622 L/min | 43.70 mL/s | Total: 0.869 L | Pump-01: ON
04:53:04.693 -> Flow: 2.622 L/min | 43.70 mL/s | Total: 0.913 L | Pump-01: ON
04:53:05.709 -> Flow: 2.597 L/min | 43.28 mL/s | Total: 0.956 L | Pump-01: ON
04:53:06.710 -> Flow: 2.622 L/min | 43.70 mL/s | Total: 1.000 L | Pump-01: ON
04:53:07.710 -> Flow: 2.622 L/min | 43.70 mL/s | Total: 1.044 L | Pump-01: ON
04:53:08.709 -> Flow: 2.622 L/min | 43.70 mL/s | Total: 1.087 L | Pump-01: ON
04:53:09.692 -> Flow: 2.597 L/min | 43.28 mL/s | Total: 1.131 L | Pump-01: ON
04:53:10.156 -> SYSTEM STOPPED
```

100% PERCENT

```
04:54:36.036 -> System: STOPPED | Float: HIGH LEVEL | Pump-01: OFF
04:54:44.704 -> HIGH LEVEL: Pump-02 ON; Pump-01 OFF.
04:54:45.701 -> Flow: 2.498 L/min | 41.64 mL/s | Total: 0.042 L | Pump-01: OFF
04:54:46.687 -> Flow: 2.671 L/min | 44.52 mL/s | Total: 0.086 L | Pump-01: OFF
04:54:47.686 -> Flow: 2.647 L/min | 44.11 mL/s | Total: 0.130 L | Pump-01: OFF
04:54:48.686 -> Flow: 2.647 L/min | 44.11 mL/s | Total: 0.174 L | Pump-01: OFF
04:54:49.703 -> Flow: 2.671 L/min | 44.52 mL/s | Total: 0.219 L | Pump-01: OFF
04:54:50.703 -> Flow: 2.647 L/min | 44.11 mL/s | Total: 0.263 L | Pump-01: OFF
04:54:51.703 -> Flow: 2.647 L/min | 44.11 mL/s | Total: 0.307 L | Pump-01: OFF
04:54:52.984 -> LEVEL NORMAL: Pump-01 restarted; Pump-02 ON.
04:54:52.984 -> Flow: 2.664 L/min | 44.39 mL/s | Total: 0.365 L | Pump-01: ON
04:54:54.002 -> Flow: 2.622 L/min | 43.70 mL/s | Total: 0.409 L | Pump-01: ON
04:54:55.002 -> Flow: 2.647 L/min | 44.11 mL/s | Total: 0.453 L | Pump-01: ON
04:54:55.985 -> Flow: 2.647 L/min | 44.11 mL/s | Total: 0.497 L | Pump-01: ON
04:54:57.002 -> Flow: 2.622 L/min | 43.70 mL/s | Total: 0.540 L | Pump-01: ON
04:54:58.002 -> Flow: 2.647 L/min | 44.11 mL/s | Total: 0.585 L | Pump-01: ON
04:54:59.002 -> Flow: 2.647 L/min | 44.11 mL/s | Total: 0.629 L | Pump-01: ON
04:54:59.981 -> Flow: 2.647 L/min | 44.11 mL/s | Total: 0.673 L | Pump-01: ON
04:55:00.998 -> Flow: 2.622 L/min | 43.70 mL/s | Total: 0.716 L | Pump-01: ON
04:55:01.985 -> Flow: 2.647 L/min | 44.11 mL/s | Total: 0.761 L | Pump-01: ON
04:55:03.000 -> Flow: 2.671 L/min | 44.52 mL/s | Total: 0.805 L | Pump-01: ON
04:55:03.985 -> Flow: 2.647 L/min | 44.11 mL/s | Total: 0.849 L | Pump-01: ON
04:55:04.984 -> Flow: 2.647 L/min | 44.11 mL/s | Total: 0.893 L | Pump-01: ON
04:55:05.986 -> Flow: 2.647 L/min | 44.11 mL/s | Total: 0.937 L | Pump-01: ON
04:55:06.986 -> Flow: 2.647 L/min | 44.11 mL/s | Total: 0.982 L | Pump-01: ON
04:55:07.985 -> Flow: 2.647 L/min | 44.11 mL/s | Total: 1.026 L | Pump-01: ON
04:55:08.984 -> Flow: 2.647 L/min | 44.11 mL/s | Total: 1.070 L | Pump-01: ON
04:55:10.001 -> Flow: 2.647 L/min | 44.11 mL/s | Total: 1.114 L | Pump-01: ON
04:55:11.001 -> SYSTEM STOPPED
```

```
22:02:39.342 -> Pump-01: controlled by float switch
22:02:39.342 -> Pump-02: RUNNING
22:02:40.032 -> [LIVE] Pulses=103490 | Frequency=165.00 Hz | Estimated flow=4.081 L/min | Estimated total volume=42.6622 L | Pump-02=100% | Float=NORMAL
22:02:41.009 -> [LIVE] Pulses=103756 | Frequency=266.00 Hz | Estimated flow=6.579 L/min | Estimated total volume=42.7719 L | Pump-02=100% | Float=NORMAL
22:02:42.022 -> [LIVE] Pulses=104023 | Frequency=267.00 Hz | Estimated flow=6.604 L/min | Estimated total volume=42.8819 L | Pump-02=100% | Float=NORMAL
22:02:43.038 -> [LIVE] Pulses=104289 | Frequency=266.00 Hz | Estimated flow=6.579 L/min | Estimated total volume=42.9916 L | Pump-02=100% | Float=NORMAL
22:02:44.008 -> [LIVE] Pulses=104556 | Frequency=267.00 Hz | Estimated flow=6.604 L/min | Estimated total volume=43.1017 L | Pump-02=100% | Float=NORMAL
22:02:45.008 -> [LIVE] Pulses=104823 | Frequency=267.00 Hz | Estimated flow=6.604 L/min | Estimated total volume=43.2117 L | Pump-02=100% | Float=NORMAL
22:02:46.008 -> [LIVE] Pulses=105090 | Frequency=267.00 Hz | Estimated flow=6.604 L/min | Estimated total volume=43.3218 L | Pump-02=100% | Float=NORMAL
22:02:47.039 -> [LIVE] Pulses=105357 | Frequency=267.00 Hz | Estimated flow=6.604 L/min | Estimated total volume=43.4319 L | Pump-02=100% | Float=NORMAL
22:02:48.016 -> [LIVE] Pulses=105625 | Frequency=268.00 Hz | Estimated flow=6.629 L/min | Estimated total volume=43.5423 L | Pump-02=100% | Float=NORMAL
22:02:49.026 -> [LIVE] Pulses=105892 | Frequency=267.00 Hz | Estimated flow=6.604 L/min | Estimated total volume=43.6524 L | Pump-02=100% | Float=NORMAL
22:02:50.018 -> [LIVE] Pulses=106160 | Frequency=268.00 Hz | Estimated flow=6.629 L/min | Estimated total volume=43.7629 L | Pump-02=100% | Float=NORMAL
22:02:51.022 -> [LIVE] Pulses=106428 | Frequency=268.00 Hz | Estimated flow=6.629 L/min | Estimated total volume=43.8734 L | Pump-02=100% | Float=NORMAL
22:02:52.042 -> [LIVE] Pulses=106697 | Frequency=269.00 Hz | Estimated flow=6.653 L/min | Estimated total volume=43.9843 L | Pump-02=100% | Float=NORMAL
22:02:53.042 -> [LIVE] Pulses=106965 | Frequency=268.00 Hz | Estimated flow=6.629 L/min | Estimated total volume=44.0947 L | Pump-02=100% | Float=NORMAL
22:02:54.018 -> [LIVE] Pulses=107234 | Frequency=269.00 Hz | Estimated flow=6.653 L/min | Estimated total volume=44.2056 L | Pump-02=100% | Float=NORMAL
22:02:55.011 -> [LIVE] Pulses=107503 | Frequency=269.00 Hz | Estimated flow=6.653 L/min | Estimated total volume=44.3165 L | Pump-02=100% | Float=NORMAL
22:02:56.008 -> [LIVE] Pulses=107771 | Frequency=268.00 Hz | Estimated flow=6.629 L/min | Estimated total volume=44.4270 L | Pump-02=100% | Float=NORMAL
22:02:57.009 -> [LIVE] Pulses=108040 | Frequency=269.00 Hz | Estimated flow=6.653 L/min | Estimated total volume=44.5379 L | Pump-02=100% | Float=NORMAL
22:02:58.048 -> [LIVE] Pulses=108309 | Frequency=269.00 Hz | Estimated flow=6.653 L/min | Estimated total volume=44.6488 L | Pump-02=100% | Float=NORMAL
22:02:59.009 -> [LIVE] Pulses=108578 | Frequency=269.00 Hz | Estimated flow=6.653 L/min | Estimated total volume=44.7597 L | Pump-02=100% | Float=NORMAL
22:03:00.008 -> [LIVE] Pulses=108847 | Frequency=269.00 Hz | Estimated flow=6.653 L/min | Estimated total volume=44.8706 L | Pump-02=100% | Float=NORMAL
22:03:01.037 -> [LIVE] Pulses=109116 | Frequency=269.00 Hz | Estimated flow=6.653 L/min | Estimated total volume=44.9814 L | Pump-02=100% | Float=NORMAL
22:03:02.055 -> [LIVE] Pulses=109385 | Frequency=269.00 Hz | Estimated flow=6.653 L/min | Estimated total volume=45.0923 L | Pump-02=100% | Float=NORMAL
22:03:02.388 ->
22:03:02.388 -> SYSTEM STOPPED
22:03:02.388 -> Pumps: OFF
```

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